

Experiment 'p_6_2_h_is_sorted_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_h_is_sorted_seq

Natural language statement: null

Method signature: p_6_2_h_is_sorted_seq(inputs: seq<int>) returns (is_sorted: bool)

Ensures

- `is_sorted == (forall i :: 0 <= i < |inputs| - 1 ==> inputs[i] <= inputs[i + 1])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_h_is_sorted_seq(inputs: seq<int>) returns (is_sorted: bool)`

The method should respect the following contract:

`ensures is_sorted == (forall i :: 0 <= i < |inputs| - 1 ==> inputs[i] <= inputs[i + 1])`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnO47XMhUEpsiln2KfAdSjNG325I2

Model: gpt-5.5-2026-04-23

Created at: 1780662951

Finish reason: stop

Usage: 298 tokens in, and 697 tokens out

```
{
  is_sorted := forall i :: 0 <= i < |inputs| - 1 ==> inputs[i] <= inputs[i + 1];
}
```

Final Program

```
method p_6_2_h_is_sorted_seq(inputs: seq<int>) returns (is_sorted: bool)
  ensures is_sorted == (forall i :: 0 <= i < |inputs| - 1 ==> inputs[i] <= inputs[i
    → + 1])
{
  is_sorted := forall i :: 0 <= i < |inputs| - 1 ==> inputs[i] <= inputs[i + 1];
}
```

Total Token Usage

Input tokens: 298

Output tokens: 697

Reasoning tokens: 642

Sum of ‘total tokens’: 995

Experiment Timings

Overall Experiment started at 1780662950268, ended at 1780662995997, lasting 45729ms (45.73 seconds)

Iteration #1 started at 1780662950286, ended at 1780662995997, lasting 45711ms (45.71 seconds)